

KETU SUNDARKUMAR PATEL

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SUMMARY

Accomplished **Senior Software Developer** with **6+ years of experience** specializing in **distributed systems, cloud infrastructure, and microservices**. Specializing in **Python, Django, Java, J2EE and Spring Framework** with expertise in **Kubernetes, AWS, Terraform, and observability tools (Prometheus, Grafana)**. Skilled in developing SOA and microservices-based platforms, migrating legacy systems, and implementing complex event processing for real-time event monitoring. **Proven ability to communicate complex technical solutions and work cross-functionally to enhance developer experience.** **Passionate about** open-source technologies, developer advocacy, and infrastructure scalability.

WORK EXPERIENCE

Aavis Pharmaceutical

Atlanta, Georgia

Senior Software Engineer

July 2024 – Current

- Adapted the migration of the REST-based APIs to a cloud-based distributed system employing AWS EC2, AWS SQS and S3 Buckets, ensuring scalability and fault tolerance.
- Optimized Oracle **SQL databases** by restructuring queries and indexing key tables, resulting in a 40% reduction in SKU management time and improving response times for large-scale transactional systems by 25%.
- Demonstrated ability to design and implement object hierarchies and inheritance structures using prototypes or ES6 classes, ensuring code reusability and minimizing duplication.
- Designed and documented high-level and low-level architectures (including UML diagrams) for a loaner project with the Promise Team, utilizing **Python, FastAPI, asyncio, and RabbitMQ** to deliver resilient and scalable microservices.
- Integrated **JSON Web Token** authentication with custom token expiry, refresh strategies, and blacklisting for secure session handling and token revocation in high-trust systems.
- Managed cloud-native services** on AWS/GCP using **Lambda, SQS, SNS, and CloudWatch**, and supported infrastructure-as-code using Terraform for secure hosted APIs and worker services.
- Designed and enforced **OAuth2.0** and OpenID Connect protocols for secure token-based authentication and delegated access, using Authlib and FastAPI Security

FeelTech Solutions Private Limited

Manhattan, New York

Software Engineer

October 2021 – June 2024

- Refactored three major legacy codebases to align with **SOLID principles**, resulting in a 40% increase in overall maintainability and enabling rapid feature development for the team within tight deadlines.
- Proficient in implementing object-oriented design patterns in **JavaScript**, utilizing concepts such as objects, classes, inheritance, and encapsulation to create scalable and maintainable codebases.
- Architected secure, scalable authentication layers leveraging **OAuth2, OpenID Connect, JWT, and RBAC**, with secure token storage, rotation, and audit logging.
- Led efforts to **automate infrastructure** using **Terraform**, reducing manual deployment time by **60%**.
- Streamlined deployment with 0 downtime using GitHub Actions **CI/CD pipelines** to ensure continuous availability.
- Explored AI/ML integration into backend components using **PyTorch, TensorFlow, scikit-learn, and FastAPI**, aligning with industry trends toward blending web and ML services.
- Implemented monitoring dashboards with Grafana and Prometheus, identifying and resolving performance bottlenecks.

Desire Infoweb Pvt Ltd.

Ahmedabad, India

Full Stack Developer

January 2018 – September 2021

- Built a **React Native application** in an agile environment, leveraging **jQuery plugins**, drag-and-drop UI components, and **JSP validation** with the **Struts Validation framework** to streamline user interactions and enhance frontend responsiveness.
- Optimized **Oracle SQL databases** by managing tables, implementing stored procedures, and tuning performance for high traffic applications. an Indian government project.
- Built and deployed **full-stack web applications** using **React.js, Django, and PostgreSQL**.
- Developed **RESTful APIs** and implemented **API security best practices**.
- Integrated **real-time monitoring tools** for logging and analytics to improve system health tracking.
- Designed and implemented **cloud-native architecture** for an **Indian government project**, improving scalability and performance.

EDUCATION

Master of Science in Computer Science
New York Institute of Technology • Manhattan, NY
Bachelor of Engineering in Information Technology
Gujarat Technological University • Gujarat, India

May 2022
GPA: 3.50/4.00
August 2020
GPA: 8.40/10.00

TECHNICAL PROJECTS

Equities Middle Office Platform - Investment Bank (Startup: Conatix)

- Developed and enhanced Java server and client-side components for a strategic Middle Office trading platform.
- Migrated multiple legacy Middle Office applications into a unified platform with SOA architecture using Java, J2EE, Spring, Hibernate, and JPA. Integrated Java Messaging Service (JMS) and Solace Messaging for communication and used Complex Event Processing (CEP) to track and monitor trade events in real time.
- Coordinated with business analysts to gather requirements and supported production releases with comprehensive testing and issue resolution.

Technologies: Java, J2EE, Spring, Hibernate, JPA, JMS, Solace, CEP, Oracle, Maven, Git, Jira, Confluence, Bitbucket, TeamCity, Sonar.

E-Financial - Internal Threat Detection System (Startup: Conatix)

- Developed a microservices-based internal threat detection system designed to analyze employee and system activities within financial institutions to identify unusual patterns that could indicate fraudulent behavior or internal threats.
- This platform will leverage real-time data analytics and machine learning to detect anomalies in transaction patterns, access logs, and other relevant data sources.
- The system will provide alerts and detailed reports to security analysts for further investigation, helping to mitigate risks associated with internal threats and safeguard sensitive financial information.

Technologies: Java, Azure ML, Azure Stream Analytics, Azure Cosmos DB

Pulse Oximeter Tracker (Research Project)

- Developed an AWS-based pulse oximeter tracker for the College of Osteopathic Medicine at NYIT.
- Collaborated with medical students to create a solution for tracking user's pulse data, improving medical diagnoses and treatment plans while demonstrating Data visualization skills.
- Successfully deployed, the solution enabled real-time pulse data monitoring for multiple users, providing more comprehensive and accurate information for medical research advancements.

Technologies: AWS, VueJS, Python

PARKED (Bachelor's Final Year Project)

- PARKED is a smart parking management system using Artificial Intelligence.
- It is used for vehicle registering, creation of invoices & management of parking.
- Uses Automatic License Plate Identification API to identify and recognize vehicles. Sold entire project INR 25K.

Technologies: Python, Django, PyCharm, OpenCV

SKILLS

Programming Languages: Java, Python, Golang, TypeScript, JavaScript, React.js

Cloud & Infrastructure: AWS (EKS, EC2, S3), GCP, Kubernetes, Terraform

Observability & Monitoring: Prometheus, Grafana, OpenTracing, ELK Stack

Tools & Technologies: Docker, CI/CD (GitHub Actions, Jenkins), PostgreSQL, MongoDB

Relevant Coursework: Algorithmic Concepts, Programming Languages, Software Engineering, Advance Machine Learning, Database Systems, Programming for Data Science, Data Visualization, Theoretical Concepts in Computers and Computation, Computer Architecture

Other: Debugging, Performance Optimization, Technical Documentation, Developer Advocacy.

Languages: English, Hindi, Gujarati